Calibrated Distributional Forecasting of LEO and 5G Link State for Prediction-Aware Multipath Transport

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***Abstract*—A terminal with low Earth orbit (LEO) satellite and terrestrial 5G paths can schedule traffic ahead of time only if it knows how each path will behave in the next seconds and how certain that knowledge is. Using 5 Hz measurements of Starlink, OneWeb, and 5G links, we forecast the round-trip time (RTT), jitter, and loss 0.2 s to 3 s ahead. First, we show that the common practice of aligning ping streams by time interpolation leaks the next sample into the model input, prove that it halves the one-step persistence error on every interpolated row, and measure a 9.7% understatement of the first-step error on Starlink. Second, we propose a recurrent encoder-decoder anchored on the last observation whose heads output a mean and five non-crossing quantiles per target and step, trained with a squared error plus pinball loss, and calibrated on held-out sessions with conformalized quantile regression. Against the same network with a scalar squared-error output, with three seeds each, the 95% interval score of the RTT forecasts improves on all six links by -8.4% to -39.3% (mean -22.0%) at a measured coverage of 0.935 to 0.954, and the RTT mean absolute error drops by -1.5% to -17.7%. The RTT root-mean-square error changes by only -4.1% to +0.4%, and a tuned linear model is within one point of the neural baseline's skill on four links. We conclude that the value of learning lies in input-dependent uncertainty that a multipath scheduler can use as a per-path risk bound.**

***Index Terms*—Conformal prediction, LEO satellite networks, link-state forecasting, multipath transport, OneWeb, quantile regression, Starlink, 5G.**

I. Introduction

LOW Earth orbit (LEO) constellations such as Starlink and OneWeb now give remote and airborne terminals broadband service where no terrestrial network reaches, and 3GPP integrates such non-terrestrial links into 5G [1]. A terminal that has both a LEO and a terrestrial path can send each packet on the path that will deliver it in time. Multipath transport protocols such as Multipath QUIC (MPQUIC) provide the mechanism [2]. Their schedulers react to smoothed RTT estimates of the recent past [3], [4]. On LEO paths this is too slow. Starlink latency jumps at globally synchronized reconfiguration instants every 15 s [5], [6]. OneWeb latency swings at satellite handovers [7]. Measured MPTCP and MPQUIC sessions over Starlink and cellular paths show scheduler failures caused by these dynamics [8]. Schedulers that use forecasts of path state instead of measurements of it have been proposed for multi-access and mobile settings [9], [10]. They need forecasts that are accurate a few hundred milliseconds to a few seconds ahead and, because a wrong forecast costs a missed deadline, they need to know how wrong each forecast may be.

Prior work on LEO performance prediction targets throughput or latency at coarse time scales or over large areas [11], [12]. Attention models have also been used to predict LEO RTT [13]. Work on terrestrial links shows that fine-grained throughput is hard to predict and that learned models should be compared with simple baselines [14], [15]. Three questions remain open for the scheduling use case. Are the measurement pipelines that produce the training data causal? Which point forecast should be reported for heavy-tailed link metrics, where a few spikes dominate the squared error? And can the uncertainty of a sub-second forecast be calibrated on sessions the model has not seen?

This paper answers these questions with measurements of a Starlink, a OneWeb, and a 5G link taken in parallel at 5 Hz on the ARA wireless living lab [16]. Its contributions are the following.

1) **A causality audit of link-state data.** We show that aligning ping streams by time interpolation, a common step when several collectors are merged on one time base, places part of the next sample in the model input. We prove that the leak reduces the one-step persistence error of every interpolated row by the interpolation weight (one half for evenly spaced rows) and measure that it hides 9.7% of the first-step RTT error on Starlink downlink, where 11.9% of the samples are interpolated. All results in this paper use causal inputs.
2) **A distributional encoder-decoder.** We extend a GRU sequence-to-sequence forecaster with additive attention [17], [18] by anchoring its decoder on the last observation and replacing each scalar head with a head that outputs a mean and five quantiles whose order is guaranteed by construction. A single loss trains the mean and the quantiles, and each output is the optimal forecast under its own scoring rule [19].
3) **Calibration on unseen sessions.** We calibrate the quantile intervals with conformalized quantile regression (CQR) [20] on held-out measurement sessions and report the coverage measured on further held-out sessions, instead of coverage computed from the test residuals.
4) **A controlled evaluation.** On six links, with three seeds, an ablation of each design step, and four non-neural references, the proposed model improves the 95% interval score of the RTT forecast by -8.4% to -39.3% and its mean absolute error by -1.5% to -17.7% against the same network with a scalar output, while the root-mean-square error stays within a few percent. We report where it does not help, including a spike-warning task on OneWeb downlink.

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Section II reviews related work. Section III describes the measurement system and formulates the forecasting problem, including the causality analysis. Section IV presents the forecaster and its calibration. Sections V and VI give the evaluation protocol and the results, and Section VII discusses their use for multipath scheduling and concludes the paper.

II. Related Work

A. LEO Network Measurements

The first measurement studies of Starlink reported median RTTs of a few tens of milliseconds with frequent spikes and loss under load [21]. Later studies traced many of the latency jumps to a global scheduler that reassigns users to satellites every 15 s [5], [6]. Long campaigns characterized the access, point-of-presence, and backbone segments [22], and Starlink throughput was shown to vary on time scales from sub-second to diurnal [23]. OneWeb has no inter-satellite links in service, and its latency changes with every satellite and gateway handover [7]. A study on the same testbed as ours measured OneWeb dynamics and the benefit of combining paths in a liquid data transport [24]. These studies explain why LEO link state changes quickly but do not forecast it at the sub-second scale needed by a packet scheduler.

B. Network Performance Prediction

Learned throughput prediction is well established for cellular links. Lumos5G maps location, mobility, and radio context to 5G throughput [25], and recurrent models predict mobile bandwidth one step ahead [26]. Classical and deep models have also been compared on cellular traces [15]. A study of sub-second cellular throughput found that learned models gain little over simple predictors at fine time scales [14]. For LEO networks, recent work predicts Starlink throughput from months of measurements [11] and global Starlink latency and throughput at coarse resolution [12]. LEO RTT has been predicted with spatiotemporal attention [13]. A concurrent preprint forecasts LEO link state with physics-informed features and quantile bands [27]. All of these report point accuracy on test data, and none checks the causality of the input alignment or calibrates uncertainty on unseen sessions. Our evaluation includes persistence, seasonal, and linear references, following the observation that simple linear forecasters are often competitive with deep sequence models [28].

C. Sequence Models and Probabilistic Forecasting

Our forecaster builds on the encoder-decoder framework [29] with gated recurrent units [17]. It uses additive attention [18] and is trained with scheduled teacher forcing [30]. Forecast accuracy is reported relative to a naive forecast, as recommended for series of different scales [31]. For probabilistic forecasts, a proper scoring rule rewards honest uncertainty [32], and the optimal point forecast depends on the loss used to evaluate it [19]. Quantile regression with the pinball loss estimates conditional quantiles directly [33]. Deep ensembles improve both accuracy and uncertainty [34].

D. Conformal Prediction

Split conformal prediction gives intervals with finite-sample coverage when calibration and test data are exchangeable [35]. CQR adapts the interval width to the input by calibrating quantile regression outputs [20]. Conformal methods have been applied to multi-horizon recurrent forecasts [36] and extended to non-exchangeable data with explicit coverage gaps [37]. Adaptive versions track distribution shift online [38]. We use CQR across measurement sessions and report the coverage achieved on separate sessions.

E. Multipath Scheduling

MPQUIC extends QUIC to several paths [2]. Its default scheduler sends on the path with the lowest smoothed RTT, a policy inherited from MPTCP [3]. BLEST and ECF account for head-of-line blocking and completion time on heterogeneous paths [4], [39]. Peekaboo learns a scheduling policy online [40]. MAMS uses mobility and handover awareness [9], and path-state forecasts have been fed to a 5G multi-access scheduler [10]. These schedulers act on point estimates. The calibrated intervals in this paper provide the risk information that they lack.

III. System Model and Problem Formulation

A. Measurement System

Fig. 1 shows the setup. A client node of the ARA wireless living lab [16] is connected to a Starlink user terminal, a OneWeb user terminal, and a 5G network through a cellular modem. In each measurement session the node pings a server at 5 Hz over the link under test, runs an iperf3 UDP flow in the downlink (DL) or uplink (UL) direction that reports throughput, jitter, and loss once per second, and logs terminal telemetry and local weather. The telemetry is link specific: dish state, satellite geometry, signal-to-noise ratio, and handover events for Starlink, signal-to-interference-plus-noise ratio (SINR), elevation, azimuth, and handover events for OneWeb, and RSRP, RSRQ, and SINR for 5G. A parser merges all collectors on the ping time base, which gives one row every $\Delta = 200$ ms. The forecaster runs next to the multipath client and supplies per-path forecasts and intervals to its scheduler.

B. Forecasting Problem

For one link, let $\mathbf{x}_t \in \mathbb{R}^F$ be the feature vector of row t and $\mathbf{y}_t \in \mathbb{R}^K$ the $K = 3$ targets: RTT (ms), iperf jitter (ms), and iperf loss (%). The targets are also part of $\mathbf{x}_t$. Given the window $\mathbf{X}_t = (\mathbf{x}_{t-L+1}, \ldots, \mathbf{x}_t)$ of $L = 30$ rows (6 s), the forecaster outputs, for every step $h \in \{1, \ldots, H\}$ with $H = 15$ (0.2–3 s) and every target k, a point forecast and a $(1-\alpha)$ prediction interval $[\ell_{t,h,k}, u_{t,h,k}]$. Sessions are the unit of independence. The model is trained on one set of sessions, selected and calibrated on a second set, and evaluated on a third.

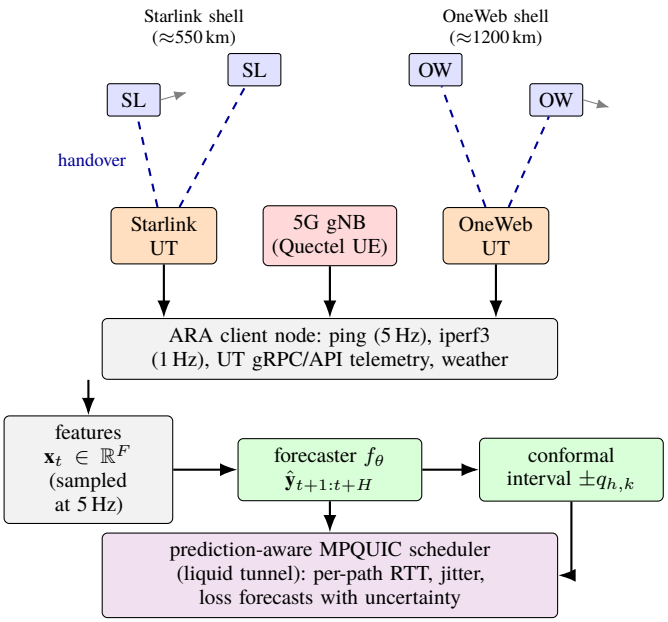


Fig. 1. Measurement system and forecasting pipeline. Three access links are measured in parallel. The forecaster maps a window of link-state features to a distribution of future RTT, jitter, and loss, and a conformal step turns it into calibrated intervals for a prediction-aware MPQUIC scheduler.

C. Scoring Rules and Optimal Forecasts

A forecast is judged by a loss $S(\hat{y}, y)$ averaged over test windows. Two losses are standard for link metrics, the squared error and the absolute error. For a $(1-\alpha)$ interval we use the interval score [32]

$$\mathrm{IS}_\alpha(\ell, u; y) = (u-\ell) + \tfrac{2}{\alpha}(\ell - y)^+ + \tfrac{2}{\alpha}(y-u)^+, \quad (1)$$

with $(z)^+ = \max\{z, 0\}$, which rewards narrow intervals and penalizes each miss in proportion to its size. The following facts guide the design of the forecaster.

Proposition 1: Let Y be the target given the input window, with finite variance. The expected squared error is minimized by $\mathbb{E}[Y]$, the expected absolute error by any median of Y, and the expected pinball loss $\rho_\tau(Y-q)$ with $\rho_\tau(e) = \max\{\tau e, (\tau-1)e\}$ by any τ-quantile of Y. The expected interval score (1) is minimized by $\ell = q_{\alpha/2}$ and $u = q_{1-\alpha/2}$.

Proof: These are standard results [19], [33]. For the interval score, $\mathrm{IS}_\alpha(\ell, u; y) = \frac{2}{\alpha}\big[\rho_{\alpha/2}(y-\ell) + \rho_{1-\alpha/2}(y-u)\big]$, which can be checked case by case for $y < \ell$, $\ell \le y \le u$, and $y > u$. The two terms are minimized separately by the stated quantiles. ∎

For heavy-tailed link metrics the mean and the median differ, and a model trained only with the squared error is not the best forecast under the absolute error. In our test sessions the RTT RMSE of the retrained baseline is 1.5 to 3.8 times its MAE, a sign of heavy tails. Proposition 1 also shows that a constant-width interval around a mean forecast cannot minimize the interval score when the spread of Y depends on the input.

D. Causality of the Input Alignment

The collectors run on independent clocks. When a ping is missing at row t, a common merging step fills it by time interpolation between the neighboring real pings. Let y_{t-1} and y_{t+1} be the real pings around a single missing row and let the parser write $\tilde{y}_t = (1-w)\, y_{t-1} + w\, y_{t+1}$ with $w \in (0,1)$ the relative time position of row t ($w = 1/2$ for evenly spaced rows). The forecast origin $\tilde{y}_t$ then contains part of the first target y_{t+1}.

Proposition 2: For a window whose last row is interpolated as above, the one-step persistence error with the interpolated origin is $y_{t+1} - \tilde{y}_t = (1-w)\,(y_{t+1} - y_{t-1})$, which is $(1-w)$ times the error with the causal origin y_{t-1}. Its squared error is therefore $(1-w)^2$ times the causal one, a factor of four for $w = 1/2$, for any realization of the process. If a fraction p of the windows have an interpolated origin with $w = 1/2$, and the causal one-step errors of these windows have the same mean square as the others, the one-step RMSE of persistence is understated by the factor $\sqrt{1 - 3p/4}$.

Proof: Substitute $\tilde{y}_t$ into $y_{t+1} - \tilde{y}_t$ for the first statement. For the second, the mean squared error over all windows is $(1-p)\,m + p\,m/4 = m(1 - 3p/4)$, where m is the causal mean squared error. ∎

The leak is not limited to persistence. Every model that reads the last input value of a target, which includes residual decoders and rolling features, can exploit it. In our Starlink DL data, 11.9% of the test RTT samples are interpolated, and Proposition 2 predicts a one-step RMSE understatement of about 5%. Replacing every interpolated input by the last real ping raises the measured one-step persistence RMSE by 9.7%, from 56.7 to 62.2 ms. The larger measured effect means the equal-error assumption of Proposition 2 does not hold: windows whose origin is interpolated have larger one-step errors than the others. In every Starlink test window whose last row is interpolated, the input value lies strictly between the previous row and the first target, against 47% of the windows with a real last ping (DL). The 5G and OneWeb RTT series contain no interpolated values. In the rest of the paper all inputs are causal and the targets are the measured series.

IV. Distributional Forecaster with Conformal Calibration

A. Encoder and Attention

Fig. 2 shows the forecaster. Each input row is concatenated with a learned embedding of the sine and cosine of the hour of day and the day of week, and a two-layer GRU encoder with hidden size $d = 128$ [17] maps the window to the states $\mathbf{e}_{t-L+1}, \ldots, \mathbf{e}_t$. At decoder step h the additive attention [18] forms the context $\mathbf{c}_h = \sum_j a_{h,j}\mathbf{e}_j$ with $a_{h,j} \propto \exp\big(\mathbf{v}^\top \tanh(\mathbf{W}_q\mathbf{s}_{h-1} + \mathbf{W}_k\mathbf{e}_j)\big)$, where $\mathbf{s}_{h-1}$ is the top decoder state. The encoder and the attention are those of the baseline. The comparison therefore isolates the decoder and the output.

B. Anchored Distributional Decoder

A two-layer GRU decoder cell takes $\mathbf{c}_h$ and an embedding of the previous mean forecast $\hat{\boldsymbol{\mu}}_{t+h-1}$, with $\hat{\boldsymbol{\mu}}_t = \mathbf{y}_t$ read from the input window, and produces $\mathbf{s}_h$. For each target

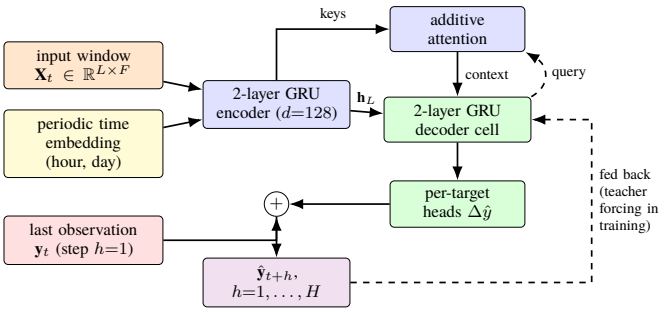


Fig. 2. Proposed forecaster. The decoder is anchored on the last observation, and each head outputs a mean and five non-crossing quantiles of the change.

k a head g_k (a two-layer perceptron) outputs six numbers $(m, r, a_1, a_2, b_1, b_2)$, from which

$$\hat{\mu}_{t+h,k} = \hat{\mu}_{t+h-1,k} + m, \qquad \hat{q}_{0.5} = \hat{\mu}_{t+h-1,k} + r, \quad (2)$$
$$\hat{q}_{0.25} = \hat{q}_{0.5} - \varsigma(a_1), \quad \hat{q}_{0.025} = \hat{q}_{0.25} - \varsigma(a_2),$$
$$\hat{q}_{0.75} = \hat{q}_{0.5} + \varsigma(b_1), \quad \hat{q}_{0.975} = \hat{q}_{0.75} + \varsigma(b_2),$$

with $\varsigma(z) = \log(1+e^z)+\epsilon$ and $\epsilon = 10^{-4}$. The anchoring in (2) makes persistence the forecast obtained when all head outputs are zero. The network only learns departures from the last value. Because $\varsigma > 0$, the five quantiles are strictly ordered for every input, which removes the quantile crossing that separate quantile heads can produce. The decoder feeds back the mean, which keeps the recursion identical to the baseline.

C. Training Loss

With targets normalized by training statistics, the loss over a batch of N windows is

$$\mathcal{L} = \frac{1}{NHK} \sum_{n,h,k} \Big[(\hat{\mu}_{nhk} - y_{nhk})^2 + \frac{1}{5} \sum_{\tau\in\mathcal{T}} \rho_\tau(y_{nhk} - \hat{q}_{\tau,nhk}) \Big], \quad (3)$$

with $\mathcal{T} = \{0.025, 0.25, 0.5, 0.75, 0.975\}$. By Proposition 1, the population minimizer of (3) over unrestricted functions of the window is the conditional mean for $\hat{\mu}$ and the conditional τ-quantile for each $\hat{q}_\tau$. These targets are compatible with the ordering constraint, since conditional quantiles are ordered. The point forecast reported for squared-error metrics is $\hat{\mu}$ and for absolute-error metrics $\hat{q}_{0.5}$. Teacher forcing is applied per sample with a probability that decays from 0.5 to 0 over half of the epochs [30].

D. Conformal Calibration Across Sessions

The learned quantiles are fitted on training sessions and may under-cover or over-cover new sessions. Let $\mathcal{C}$ be the windows of the calibration sessions and, for each step h and target k, define the CQR score [20]

$$s_i = \max\{\hat{q}_{\alpha/2}(\mathbf{X}_i) - y_i,\ y_i - \hat{q}_{1-\alpha/2}(\mathbf{X}_i)\}, \qquad i \in \mathcal{C}. \quad (4)$$

With $c_{h,k}$ the $\lceil (|\mathcal{C}|+1)(1-\alpha) \rceil$-th smallest score, the interval is $[\hat{q}_{\alpha/2} - c_{h,k},\ \hat{q}_{1-\alpha/2} + c_{h,k}]$. If the calibration and test windows are exchangeable, its coverage is at least $1-\alpha$ [20]. Windows of one session overlap and sessions differ in time of day and load. Exchangeability therefore holds only approximately [37],

Algorithm 1 Training, calibration, and online use

Require: training, calibration, and test sessions, α, seeds $\mathcal{S}$

1: **for** each seed in $\mathcal{S}$ **do**
2: train the forecaster with (3) on training sessions, select the checkpoint with the lowest smoothed calibration-session loss
3: **end for**
4: average the means and quantiles of the seeds
5: **for** each step h and target k **do**
6: compute the CQR scores on calibration windows and the offset $c_{h,k}$
7: **end for**
8: **online:** for a new window, output $\hat{\mu}$, $\hat{q}_{0.5}$, and $[\hat{q}_{\alpha/2} - c_{h,k}, \hat{q}_{1-\alpha/2} + c_{h,k}]$ for all h, k

TABLE I
DATASETS (5 HZ ROWS, WINDOWS OF 30 INPUT AND 15 TARGET ROWS).

Link	Sessions	Train	Calib.	Test	Features
OneWeb DL	45/9/11	117,558	22,655	27,492	9/31
OneWeb UL	45/9/11	115,847	22,751	28,125	9/31
Starlink DL	37/7/9	105,906	20,025	25,838	19/41
Starlink UL	37/7/9	106,165	20,090	25,666	19/41
5G DL	266/57/58	194,725	166,266	169,204	10/32
5G UL	265/56/58	193,502	162,894	168,519	10/32

and we report the coverage measured on test sessions. For a model without quantiles we use the split conformal interval $\hat{y} \pm c$ with c the same order statistic of $|y_i - \hat{y}_i|$ [35].

E. Seed Ensembles and Complexity

Three models trained with different seeds are combined by averaging their mean and each quantile, which keeps the ordering, and the ensemble is calibrated in the same way [34]. The forecaster has 468,511 parameters against 449,592 for the scalar baseline on OneWeb DL. A forecast needs one encoder pass and H decoder steps, as in the baseline. Algorithm 1 summarizes training, calibration, and inference.

V. Evaluation Setup

A. Data

Table I lists the six links. Sessions are split at random into 70% training, 15% calibration (also used for model selection), and 15% test sessions with a fixed seed. All windows of a session stay in one set. The OneWeb UL jitter and loss are zero in every row of every session. They are kept as targets for uniformity but excluded from all summaries. The 5G sets have about seven times more windows than the satellite sets, and every fourth training window is used for all methods. Overlapping windows are highly redundant, and the subsampling keeps the study within one laptop GPU.

B. Inputs

The base feature set of each link (left number in the last column of Table I) holds the three targets, the iperf throughput, the link telemetry listed in Section III, and for Starlink and

5G three weather variables. The extended set adds, for each target, seven causal statistics computed within the session: the number of rows since the value last changed (the phase in the report cycle, since jitter and loss are one-second reports held for five rows), the mean and maximum of the last second, the means of the last 30 s and 60 s, the standard deviation of the last 60 s, and the session-to-date mean. It also adds the time since the previous row. Radio values outside physical ranges, such as the logger sentinel -32768 in some 5G SINR records, are treated as missing, and non-target inputs are clipped to ± 5 standard deviations after normalization with training statistics.

C. Methods

- **Persistence:** $\hat{\mathbf{y}}_{t+h} = \mathbf{y}_t$.
- **Seasonal:** $\hat{\mathbf{y}}_{t+h} = \mathbf{y}_{t+h-5\lceil h/5 \rceil}$, the value one second earlier.
- **Ridge** and **Ridge-E:** a linear model per target and step that predicts $\mathbf{y}_{t+h} - \mathbf{y}_t$ from the last input row and the 30-row target history, on the base and extended features. The penalty is chosen on the calibration sessions from $\{10^{-2}, \ldots, 10^{4}\}$.
- **S2S:** the GRU encoder-decoder of Section IV with one scalar head per target, a decoder started from zero, and the squared-error loss, on the base features. This is the model used in our earlier deployment of the prediction-aware scheduler, retrained here on causal inputs.
- **Ablation steps:** S2S-A anchors the decoder as in (2). S2S-AE also uses the extended features. **DS2S** adds the distributional heads and the loss (3). DS2S-E is the three-seed ensemble of DS2S, and S2S-E that of S2S.

All neural models use AdamW with learning rate and weight decay 10^{-3}, batch size 512, gradient clipping at norm 5, and a plateau schedule that halves the learning rate after 8 epochs without improvement. The checkpoint with the lowest exponentially smoothed calibration-session loss (factor 0.3) is kept, and training stops after 15 epochs without improvement or at 60 epochs. All 72 runs stopped early, after 16 to 44 epochs. Each neural method is trained with three seeds on an Apple M3 Max GPU.

D. Metrics

For each target and step we compute the MAE (using $\hat{q}_{0.5}$ for DS2S), the RMSE and R^2 (using $\hat{\mu}$), the RMSE skill $1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{pers}}$ against persistence [31], the empirical coverage (PICP) of the calibrated 95% interval, and the interval score (1). We report averages over the 15 steps, means and standard deviations over seeds, and relative changes against S2S.

VI. Results

A. RTT Accuracy and Uncertainty

Table II gives the RTT results per link. Against S2S, DS2S lowers the interval score on all six links by -8.4% to -39.3% (mean -22.0%) and the MAE by -1.5% to -17.7% (mean -7.5%). The RMSE changes by -4.1% to +0.4% (mean -1.2%), which is within the seed spread on most links. With three-seed ensembles on both sides, the interval score improves by -9.4% to -41.1% and the MAE by -1.6% to -17.8%. The largest MAE gains are on 5G UL (-17.7%), Starlink DL (-10.9%), and OneWeb DL (-9.8%). By Proposition 1, the median is the optimal absolute-error forecast, and it differs from the mean learned by the scalar model whenever the conditional distribution is skewed.

TABLE II
RTT on the test sessions, averaged over 0.2–3 s. Mean ± std over three seeds and three-seed ensembles (-E). MAE, RMSE, and IS in ms. IS and PICP refer to the calibrated 95% interval.

Link	Model	MAE	RMSE	R^2	IS	PICP
OneWeb DL	S2S	17.99 ± 1.50	68.93 ± 1.26	0.175 ± 0.028	329.4 ± 5.4	0.953 ± 0.001
	DS2S	16.22 ± 0.03	67.38 ± 0.50	0.212 ± 0.013	266.7 ± 3.6	0.949 ± 0.006
	S2S-E	17.75	68.16	0.192	323.4	0.953
	DS2S-E	15.94	66.61	0.230	259.0	0.949
OneWeb UL	S2S	71.83 ± 0.88	150.16 ± 0.86	0.746 ± 0.003	1059.8 ± 8.7	0.952 ± 0.000
	DS2S	70.10 ± 1.66	150.82 ± 5.08	0.743 ± 0.017	804.9 ± 19.4	0.950 ± 0.001
	S2S-E	68.76	145.43	0.762	1026.7	0.951
	DS2S-E	67.51	145.35	0.761	756.9	0.950
Starlink DL	S2S	22.32 ± 0.57	37.72 ± 0.29	0.632 ± 0.006	246.7 ± 3.1	0.950 ± 0.000
	DS2S	19.89 ± 0.58	36.19 ± 0.17	0.661 ± 0.003	149.6 ± 1.3	0.954 ± 0.002
	S2S-E	21.60	37.04	0.645	243.5	0.950
	DS2S-E	18.45	35.20	0.679	143.4	0.953
Starlink UL	S2S	3.91 ± 0.07	7.07 ± 0.03	0.351 ± 0.005	38.4 ± 0.1	0.932 ± 0.001
	DS2S	3.80 ± 0.02	7.08 ± 0.03	0.350 ± 0.005	35.2 ± 1.7	0.935 ± 0.009
	S2S-E	3.88	7.04	0.357	38.2	0.930
	DS2S-E	3.76	7.04	0.358	34.6	0.936
5G DL	S2S	3.20 ± 0.00	4.64 ± 0.01	0.231 ± 0.004	25.3 ± 0.2	0.953 ± 0.001
	DS2S	3.15 ± 0.01	4.59 ± 0.01	0.246 ± 0.002	22.0 ± 0.0	0.950 ± 0.001
	S2S-E	3.18	4.61	0.239	25.2	0.953
	DS2S-E	3.12	4.57	0.254	21.7	0.950
5G UL	S2S	12.36 ± 0.08	19.90 ± 0.00	0.016 ± 0.000	127.9 ± 0.5	0.948 ± 0.000
	DS2S	10.18 ± 0.01	19.84 ± 0.00	0.022 ± 0.000	92.0 ± 0.1	0.949 ± 0.001
	S2S-E	12.35	19.89	0.017	127.9	0.948
	DS2S-E	10.15	19.83	0.023	91.7	0.950

Both models reach the nominal coverage after calibration. The measured RTT coverage of DS2S is 0.935 to 0.954 across links (S2S: 0.932 to 0.953), and Fig. 3 shows that it holds at every step. The coverage is lowest on Starlink UL for both models, which points to a difference between the calibration and test sessions of that link. The gain in interval score therefore comes from width, not from coverage. Fig. 4 shows why. The DS2S interval widens before and during RTT bursts and narrows in quiet periods, while a constant-width interval must be as wide as the bursts at all times to reach the same coverage.

The evaluation protocol decides whether this gain is visible. Table III scores the same models with two kinds of 95% intervals. The first takes the empirical 2.5% and 97.5% quantiles of the test residuals around the mean forecast, a common practice that yields 95% coverage by construction. The second is the calibrated interval of Section IV. Under the first protocol DS2S and S2S are within 2% of each other on every link, because both intervals have a constant width and the mean forecasts are similar. Under the calibrated protocol DS2S is 8% to 39% better. Test-residual intervals hide the value of an input-dependent width, and they also overstate the coverage that can be expected in operation.

Fig. 5 shows the MAE and the interval score per step. On Starlink DL the persistence error is not monotone in the horizon. It is lower at 1 s and 2 s than at 0.2 s, because the

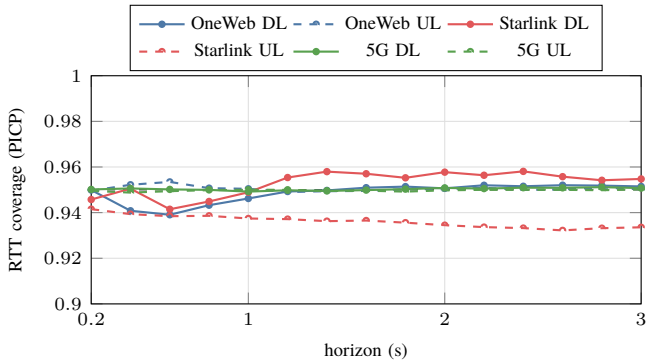


Fig. 3. Measured coverage of the calibrated 95% RTT interval of DS2S-E on the test sessions, per horizon step.

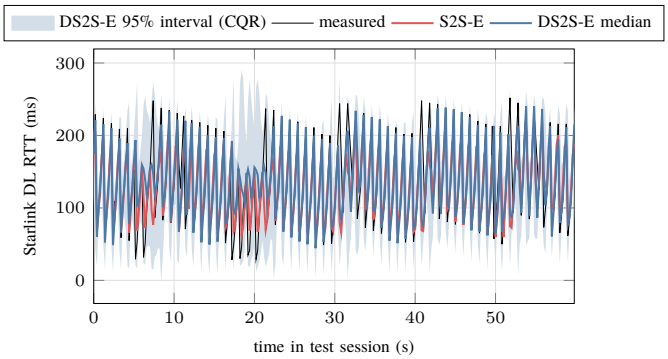


Fig. 4. One minute of a Starlink DL test session, forecast 1 s ahead ($h = 5$), with the calibrated DS2S-E interval.

downlink RTT under the 1 Hz iperf load has a one-second pattern. The seasonal reference captures this pattern alone and reaches 28% RMSE skill on that link (Table V). The learned models absorb the pattern, and their errors grow smoothly with the horizon. The DS2S gain in MAE and interval score is present at every step on all links.

B. Ablation

Table IV adds the design steps one at a time. Anchoring the decoder (S2S-A) and adding the extended features (S2S-AE) do not lower the MAE, and on OneWeb DL they raise it by 9.8% and 19.0%. A squared-error model anchored on a spiky last value follows the spikes. The extended features do carry information about the conditional mean, and S2S-AE has the lowest RMSE of all single models on three links. The MAE and interval-score gains appear only with the distributional heads, and the ensemble adds up to 6.4 points on Starlink DL. The design choice that matters is therefore the output, not the decoder initialization or the features.

C. Comparison with Simple References

Table V compares all methods. Persistence is a poor RTT forecast at 5 Hz because single pings are noisy, and every neural model and Ridge-E has a positive skill against it on every link. Ridge-E is within one point of the S2S skill on OneWeb DL, both Starlink links, and 5G UL, in line with the finding that linear forecasters are strong references [28]. Ridge on the base features fails on 5G UL (-43% skill) because of unremoved logger sentinels, which the extended pipeline treats as missing. DS2S-E has the lowest MAE on every link and the highest skill on every link, tied with Ridge-E and DS2S on 5G UL. On RMSE skill, the margin of DS2S-E over the best linear model ranges from zero (5G UL) to 9.8 points (OneWeb UL).

TABLE III
RTT INTERVAL SCORE (MS, SEED MEANS) WITH TEST-RESIDUAL INTERVALS AROUND THE MEAN (LEFT) AND WITH CALIBRATED INTERVALS (RIGHT).

	Test-residual interval			Calibrated interval		
Link	S2S	DS2S	Δ%	S2S	DS2S	Δ%
OneWeb DL	309.1	315.6	+2.1	329.4	266.7	-19.0
OneWeb UL	1043.6	1039.8	-0.4	1059.8	804.9	-24.0
Starlink DL	245.9	243.5	-1.0	246.7	149.6	-39.3
Starlink UL	36.8	37.3	+1.2	38.4	35.2	-8.4
5G DL	24.7	24.4	-1.5	25.3	22.0	-13.3
5G UL	97.3	97.5	+0.2	127.9	92.0	-28.1

D. Significance and Adaptivity

Windows of one session are correlated. We therefore also compare the ensembles session by session (Table VI). DS2S-E has the lower RTT MAE in 142 of the 156 test sessions, and a two-sided Wilcoxon signed-rank test over sessions gives $p < 0.01$ on five links. OneWeb UL, with 11 sessions and 8 wins, is not significant ($p = 0.1475$). The same table gives the Spearman correlation between the width of the DS2S-E 95% interval and the absolute error of its median over all test windows. It is positive on every link, from 0.17 on Starlink UL to 0.64 on Starlink DL, which confirms that the interval widens when the forecast is likely to be wrong. The adaptivity is largest on the links whose RTT alternates between calm periods and bursts.

E. Jitter and Loss

Jitter and loss are one-second iperf reports with long runs of zeros and rare large values. For these targets the point accuracy of DS2S and S2S is similar, with differences in both directions: the loss MAE of DS2S is higher on OneWeb DL and lower on Starlink DL and both 5G links. The calibrated interval score of DS2S is lower for both targets on all five links where they vary, by a factor of up to three (Starlink DL jitter: 2.8 against 8.5 ms). Table VII lists the values.

F. RTT Spike Warning

A scheduler may only need to know whether a path is about to degrade. We label a spike when the RTT at $t + h$ exceeds the 90th percentile of the calibration sessions and score each method by the area under the ROC curve (AUROC), using the upper quantile $\hat{q}_{0.975}$ of DS2S-E and the mean of S2S-E. Averaged over the steps, DS2S-E is the best warning signal on Starlink DL (AUROC 0.901 against 0.832), slightly better on Starlink UL (0.809 against 0.797), 5G DL, and 5G UL, about equal on OneWeb UL, and worse on OneWeb DL (0.732

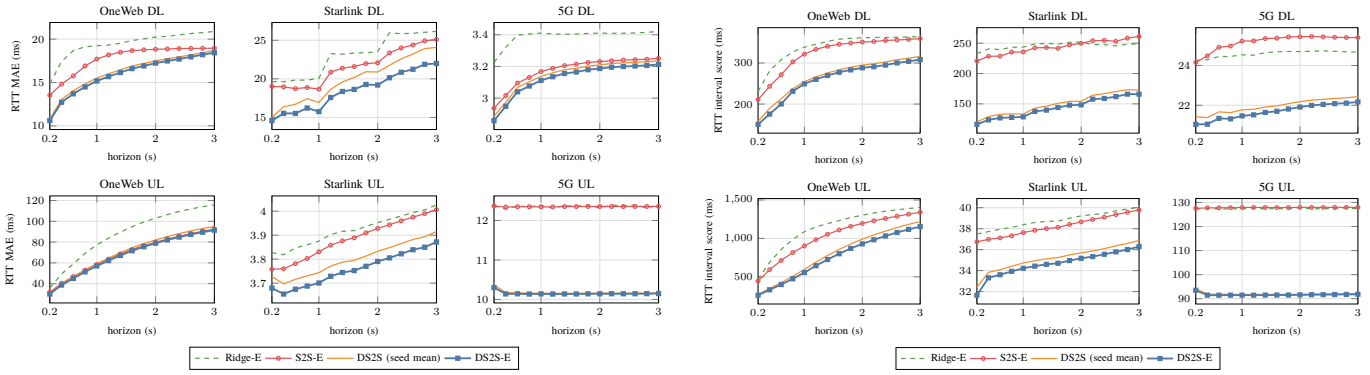


Fig. 5. RTT MAE (left) and calibrated 95% interval score (right) per horizon step on the six links. Lower is better.

TABLE IV
ABLATION: CHANGE (%) OF THE RTT MAE, RMSE, AND CALIBRATED INTERVAL SCORE AGAINST S2S (SEED MEANS, DS2S-E AGAINST THE S2S SEED MEAN). NEGATIVE IS BETTER.

	OneWeb DL			OneWeb UL			Starlink DL			Starlink UL			5G DL			5G UL		
Model	MAE	RMSE	IS	MAE	RMSE	IS	MAE	RMSE	IS	MAE	RMSE	IS	MAE	RMSE	IS	MAE	RMSE	IS
S2S-A	+9.8	-0.9	-1.1	+2.7	+3.3	+4.5	+0.7	-0.1	-1.1	+0.2	+0.2	+0.7	+0.9	+0.2	-0.7	+0.7	+0.1	-0.1
S2S-AE	+19.0	-2.8	-1.5	-1.2	-2.5	-2.6	-7.8	-4.5	-1.9	-0.7	+0.1	+0.9	+0.6	-0.6	-2.4	+0.1	-0.3	-0.5
DS2S	-9.8	-2.2	-19.0	-2.4	+0.4	-24.0	-10.9	-4.1	-39.3	-2.8	+0.1	-8.4	-1.5	-1.0	-13.3	-17.7	-0.3	-28.1
DS2S-E	-11.4	-3.4	-21.4	-6.0	-3.2	-28.6	-17.3	-6.7	-41.9	-3.9	-0.5	-9.8	-2.2	-1.6	-14.4	-17.8	-0.4	-28.3

TABLE V
RTT MAE (MS) AND RMSE SKILL AGAINST PERSISTENCE (%) OF ALL METHODS (SEED MEANS FOR THE NEURAL MODELS).

	OneWeb DL		OneWeb UL		Starlink DL		Starlink UL		5G DL		5G UL	
Method	MAE	skill	MAE	skill	MAE	skill	MAE	skill	MAE	skill	MAE	skill
Persistence	21.23	0.0	89.14	0.0	44.56	0.0	5.16	0.0	4.90	0.0	15.83	0.0
Seasonal	22.59	-8.7	101.0	-12.4	21.70	28.3	5.18	-0.9	5.03	-1.8	15.93	-0.2
Ridge	20.96	21.3	98.38	8.6	23.43	40.5	3.98	25.1	3.42	28.6	12.59	-43.3
Ridge-E	19.42	22.5	87.93	13.0	23.04	41.7	3.92	25.9	3.39	29.5	12.36	29.1
S2S	17.99	22.7	71.83	20.2	22.32	41.4	3.91	26.3	3.20	30.8	12.36	28.8
DS2S	16.22	24.5	70.10	19.8	19.89	43.8	3.80	26.2	3.15	31.5	10.18	29.1
DS2S-E	15.94	25.3	67.51	22.8	18.45	45.3	3.76	26.6	3.12	31.9	10.15	29.1

TABLE VI
PER-SESSION COMPARISON OF THE RTT MAE (DS2S-E AGAINST S2S-E) AND ADAPTIVITY OF THE DS2S-E 95% INTERVAL.

Link	Sessions	Wins	Median Δ (%)	Wilcoxon p	ρ(width, $\lvert e \rvert$)
OneWeb DL	11	11	-9.3	0.0010	0.43
OneWeb UL	11	8	-2.2	0.1475	0.50
Starlink DL	9	8	-15.6	0.0078	0.64
Starlink UL	9	9	-2.6	0.0039	0.17
5G DL	58	48	-1.6	$< 10^{-4}$	0.19
5G UL	58	58	-18.5	$< 10^{-4}$	0.23

against 0.765). OneWeb DL spikes are short and isolated, and a mean forecast that follows the last value reacts to them faster than an upper quantile that is already wide. Fig. 6 shows the AUROC per step.

G. Sensitivity to the Design Settings

We repeated the comparison on the four satellite links with two seeds for a 3 s and a 12 s input window and a 6 s horizon, and varied the hidden size (64, 256) and the pinball weight in (3) (0.25, 4) for DS2S. Table VIII shows that the advantage of DS2S over S2S holds in every window and horizon setting: the interval score improves by 7% to 41% and the MAE improves on every link, while the RMSE changes by -7.8% to $+2.1\%$. For DS2S itself, a pinball weight of 4 lowers the MAE and the interval score on all four links (OneWeb DL MAE -10.4%) at an RMSE change of at most $+0.4\%$, which indicates that the main results with weight 1 understate what the model can reach. The best input window depends on the link: 12 s lowers the OneWeb UL MAE by 9.4% and raises the OneWeb DL MAE by 11.2%.

H. Robustness and Cost

About 11.9% of the Starlink DL test targets are interpolated by the parser (Section III-D). Restricted to the measured targets, the RTT MAE is 17.86 ms for DS2S-E and 21.25 ms for S2S-E, against 18.45 and 21.60 ms on all targets. The comparison does not depend on them. The selected checkpoints were reached after at most 29 epochs in all runs, well inside the epoch budget. On one CPU thread of the laptop, one forecast

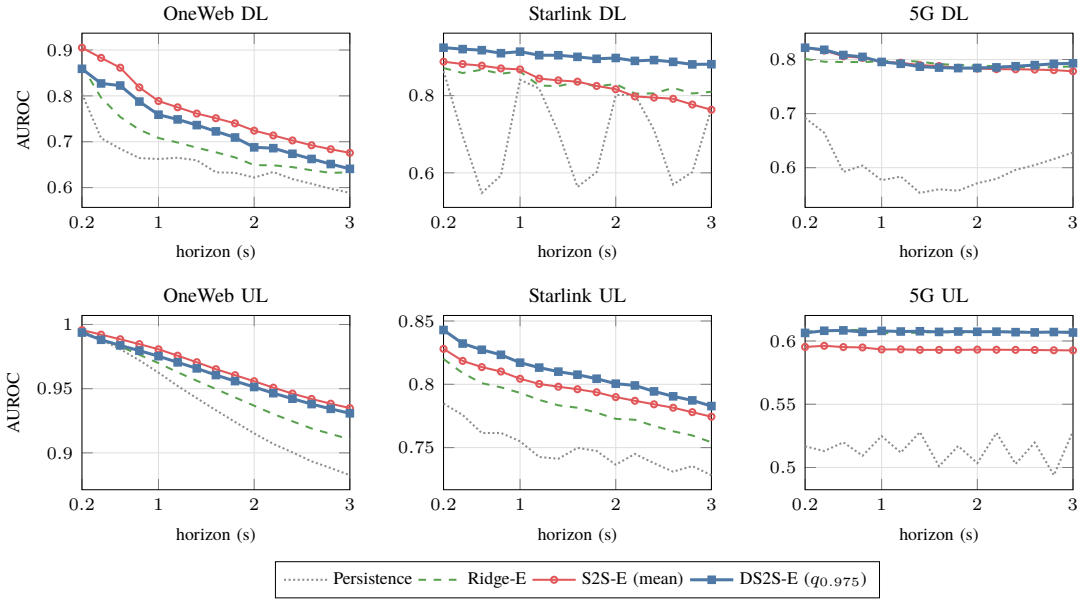


Fig. 6. RTT spike warning: AUROC for detecting that the RTT at $t+h$ exceeds the 90th percentile of the calibration sessions. DS2S-E scores with its upper quantile, the others with their point forecast.

TABLE VII
JITTER (MS) AND LOSS (%) ON THE TEST SESSIONS (SEED MEANS, ONEWEB UL OMITTED SINCE BOTH ARE ALWAYS ZERO). IS USES THE CALIBRATED 95% INTERVAL.

		Jitter			Loss		
Link	Model	MAE	RMSE	IS	MAE	RMSE	IS
OneWeb DL	S2S	0.093	0.351	1.089	0.372	2.926	9.712
	DS2S	0.095	0.353	1.020	0.458	2.939	6.968
	DS2S-E	0.093	0.351	0.991	0.419	2.918	6.517
Starlink DL	S2S	0.253	1.121	8.527	2.948	6.552	42.780
	DS2S	0.263	1.052	2.796	2.649	6.314	31.769
	DS2S-E	0.238	1.018	2.471	2.618	6.270	31.348
Starlink UL	S2S	0.176	0.332	1.772	0.330	2.152	8.132
	DS2S	0.171	0.332	1.608	0.332	2.173	6.712
	DS2S-E	0.169	0.328	1.580	0.298	2.104	5.978
5G DL	S2S	0.034	0.062	0.385	0.068	0.265	1.771
	DS2S	0.033	0.061	0.277	0.048	0.229	0.601
	DS2S-E	0.032	0.061	0.271	0.044	0.221	0.551
5G UL	S2S	0.393	1.052	6.918	0.520	2.765	15.250
	DS2S	0.333	1.046	5.226	0.399	2.650	10.050
	DS2S-E	0.330	1.044	5.190	0.387	2.623	9.737

of all 15 steps and three targets takes 10.0 ms for DS2S and 6.9 ms for S2S, far below the 200 ms between measurements.

VII. Discussion and Conclusion

A. Use in a Multipath Scheduler

The calibrated upper bound $\hat{q}_{0.975} + c_{h,k}$ is a per-path RTT bound that held about 95% of the time on unseen sessions. A deadline-aware scheduler can send a packet on a path whose bound at the packet's delivery horizon is below the deadline, and duplicate it on two paths only when no single bound qualifies. Because the DS2S bound follows the input, it is tight on a calm path and loose only around bursts (Fig. 4). A constant-width bound around a mean forecast would declare a path unsafe more often for the same coverage. This is the practical meaning of the lower interval score in Table II. The median forecast suits the ordering of paths by expected delay, where a single spike should not dominate, and the mean suits throughput and capacity estimates, where the squared error is the right loss. Both come from one forward pass.

B. Lessons for Link-State Forecasting

Three lessons apply beyond our testbed. First, measurement pipelines need a causality check. A time interpolation that is harmless for plots places future samples in the inputs, and its effect is concentrated at the shortest horizons, where schedulers operate. Second, at sub-second resolution the squared error of a point forecast is hard to improve. None of our changes reduced the RTT RMSE by more than 7% on any link, and a tuned linear model came within one point of the neural baseline's skill on four links. Claims of large point-accuracy gains at this scale should be checked against such references [14], [28]. Third, the gain of learning is in the shape of the predictive distribution, which only shows when intervals are calibrated on held-out sessions. Calibrating on the test residuals, as is common, gives the nominal coverage by construction and hides the difference.

C. Limitations

The sessions were recorded at one site, with one ping server and constant-rate iperf traffic, and the link conditions of a moving or airborne terminal may differ. The calibration sessions also serve for model selection and sessions are not exchangeable. The coverage is therefore measured rather than guaranteed. Adaptive conformal updates [38] can track a drifting offset online. The iperf reports are stamped at the start of their one-second interval, which may let jitter and loss inputs overlap the first second of the horizon for all methods

TABLE VIII
CHANGE (%) OF DS2S AGAINST S2S TRAINED WITH THE SAME INPUT WINDOW AND HORIZON (RTT, TWO SEEDS, ALL HORIZON STEPS OF THE SETTING). NEGATIVE IS BETTER.

	OneWeb DL			OneWeb UL			Starlink DL			Starlink UL		
Setting	MAE	RMSE	IS	MAE	RMSE	IS	MAE	RMSE	IS	MAE	RMSE	IS
default (6 s input, 3 s horizon)	-13.7	-3.0	-18.6	-1.1	+2.1	-23.3	-11.8	-4.1	-39.0	-3.7	-0.3	-7.2
input 15 rows (3 s)	-5.4	-1.4	-18.9	-11.1	-7.8	-29.9	-10.0	-4.1	-41.1	-1.8	-0.6	-12.3
input 60 rows (12 s)	-3.6	-2.1	-19.0	-11.4	-6.2	-27.2	-10.3	-2.4	-37.2	-1.5	-1.3	-12.5
horizon 30 steps (6 s)	-6.3	-0.7	-10.4	-4.6	-1.4	-14.7	-0.2	-2.1	-35.6	-2.2	+0.2	-10.4

alike. The scheduling benefit is argued from forecast quality, and an end-to-end evaluation inside the MPQUIC client is the next step.

D. Conclusion

We studied short-horizon forecasting of RTT, jitter, and loss for Starlink, OneWeb, and 5G links at 5 Hz. We showed that interpolation-based alignment leaks future samples into the inputs and quantified the effect. We proposed an anchored GRU encoder-decoder with non-crossing distributional heads, trained with a squared error plus pinball loss and calibrated across sessions with CQR. On six links and three seeds it improved the 95% interval score of the RTT forecasts by -8.4% to -39.3% and the MAE by -1.5% to -17.7% against the same network with a scalar output, at a measured coverage of 0.935 to 0.954 and with an unchanged RMSE. Future work will close the loop with the multipath scheduler, add the state of the other paths as inputs, and adapt the calibration online.

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